

Probability and Statistics

Exercise 4 - Zoe Zhou

Zoe Zhou

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- 1 Basic Concepts
- 2 How to find PMF: Exercise 1
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- **Random Variables (R.V.)**
- **Discrete Random Variables**
- **Probability Mass Function (PMF)**
- **Independent Random Variables**
- **Binomial Distribution**

Exercise Today

Section 3.3: Exercises 1, 2, 3, 4, and 5

Definition

A **random variable** is a function $X : S \rightarrow \mathbb{R}$ that maps each outcome in the sample space S to a real number.

- **Notation:** Random variables are typically denoted by uppercase letters: $X, Y, Z, \dots$
- **Range:** The set of all possible values of X is denoted by $\text{Range}(X)$ or R_X .

Example: Tossing a coin twice. Let X be the number of Heads.

- $S = \{TT, TH, HT, HH\}$
- $X(TT) = 0, X(TH) = 1, X(HT) = 1, X(HH) = 2$

Definition

A random variable is **discrete** if its range R_X is **countable** (i.e., either finite or countably infinite).

Examples:

- ① **Finite:** The result of a die roll, $R_X = \{1, 2, 3, 4, 5, 6\}$.
- ② **Countably Infinite:** The number of tosses until the first Head appears, $R_X = \{1, 2, 3, \dots\}$.

Probability Mass Function (PMF)

The probability distribution of a **Discrete Random Variable** is characterized by its PMF.

Definition

Let X be a discrete R.V. with range $R_X = \{x_1, x_2, \dots\}$. The function

$$P_X(x_k) = P(X = x_k)$$

for $k = 1, 2, \dots$ is called the **Probability Mass Function (PMF)** of X .

Note: We often write $P(X = x)$ instead of $P(\{s \in S : X(s) = x\})$.

Properties of PMF

A valid PMF must satisfy:

- 1 **Non-negativity:** $0 \leq P_X(x) \leq 1$ for all x .
- 2 **Normalization:** $\sum_{x \in R_X} P_X(x) = 1$.
- 3 **Subset Probability:** For any set $A \subset R_X$:

$$P(X \in A) = \sum_{x \in A} P_X(x)$$

In plain words...

- **No magic numbers:** Each individual probability must be a valid percentage between 0% and 100%. Negative probability doesn't exist!
- **No missing outcomes:** If you add up the chances of every possible thing that could happen, they must account for the entire "pie" (100%).
- **Simple addition:** To find the probability of a "range" or "category" (like X being an even number), just sum up the individual weights of each outcome in that group.

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Exercise 1: Problem

1. Let X be a discrete random variable with the following PMF

$$P_X(x) = \begin{cases} \frac{1}{2} & \text{for } x = 0 \\ \frac{1}{3} & \text{for } x = 1 \\ \frac{1}{6} & \text{for } x = 2 \\ 0 & \text{otherwise} \end{cases}$$

- (a) Find R_X , the range of the random variable X .
- (b) Find $P(X \geq 1.5)$.
- (c) Find $P(0 < X < 2)$.
- (d) Find $P(X = 0|X < 2)$

Exercise 1: Range of R.V.

(a) Find R_X (the range of X)

- The range R_X consists of all possible values x for which $P_X(x) > 0$.
- From the PMF, we see the non-zero probabilities occur at $x = 0, 1, 2$.
- Therefore: $R_X = \{0, 1, 2\}$.

Exercise 1: Interval Probabilities

(b) Find $P(X \geq 1.5)$

- In our range $\{0, 1, 2\}$, only $x = 2$ satisfies $x \geq 1.5$.
- $P(X \geq 1.5) = P(X = 2) = P_X(2) = \mathbf{1/6}$.

(c) Find $P(0 < X < 2)$

- Only $x = 1$ satisfies the strict inequality $0 < x < 2$.
- $P(0 < X < 2) = P(X = 1) = P_X(1) = \mathbf{1/3}$.

Exercise 1: Conditional Probability

(d) Find $P(X = 0 \mid X < 2)$

Recall: Conditional Probability Formula

$$P(A|B) = \frac{P(A \cap B)}{P(B)}$$

Meaning: We "zoom in" on the outcomes where B is true.

Step-by-step Calculation:

- ❶ **Numerator:** $P(X = 0 \text{ and } X < 2) = P(X = 0) = 1/2$.
- ❷ **Denominator:** $P(X < 2) = P_X(0) + P_X(1) = 1/2 + 1/3 = 5/6$.
- ❸ **Result:**

$$P(X = 0 \mid X < 2) = \frac{1/2}{5/6} = \frac{1}{2} \times \frac{6}{5} = \mathbf{3/5}$$

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Exercise 2: Problem

2. Let X be the number of the cars being repaired at a repair shop. We have the following information:
- At any time, there are at most 3 cars being repaired.
 - The probability of having 2 cars at the shop is the same as the probability of having one car.
 - The probability of having no car at the shop is the same as the probability of having 3 cars.
 - The probability of having 1 or 2 cars is half of the probability of having 0 or 3 cars.

Find the PMF of X .

Exercise 2: From Logic to Equations

Problem: X is the number of cars repaired at a shop.

- At most 3 cars: $R_X = \{0, 1, 2, 3\}$.
- $P(X = 2) = P(X = 1)$.
- $P(X = 0) = P(X = 3)$.
- $P(X = 1 \text{ or } 2) = \frac{1}{2}P(X = 0 \text{ or } 3)$.

Goal

Find the PMF of X .

Exercise 2: Setting up Variables

Let's simplify by setting $P_X(1) = \alpha$.

① From the 2nd bullet: $P_X(2) = P_X(1) = \alpha$.

② From the 4th bullet:

$$P_X(1) + P_X(2) = \frac{1}{2} [P_X(0) + P_X(3)]$$

Substitute $P_X(1)$ and $P_X(2)$ with α :

$$\alpha + \alpha = \frac{1}{2} [2 \cdot P_X(0)] \implies 2\alpha = P_X(0)$$

③ From the 3rd bullet: $P_X(3) = P_X(0) = 2\alpha$.

Summary so far:

$$P_X(1) = P_X(2) = \alpha \text{ and } P_X(0) = P_X(3) = 2\alpha.$$

Exercise 2: Solving for α

We use the **Normalization Property** ($\sum P_X(k) = 1$):

$$P_X(0) + P_X(1) + P_X(2) + P_X(3) = 1$$

$$2\alpha + \alpha + \alpha + 2\alpha = 1$$

$$6\alpha = 1 \implies \alpha = 1/6$$

Final PMF Result

$$P_X(k) = \begin{cases} 1/3 & k = 0, 3 \quad (\text{since } 2\alpha = 2/6) \\ 1/6 & k = 1, 2 \\ 0 & \text{otherwise} \end{cases}$$

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Exercise 3: Problem

Problem Statement

Roll two fair dice. Let X and Y be the numbers shown. Find the range and PMF of $Z = X - Y$.

Exercise 3: Difference of Two Dice

Key Tool: Independence

X, Y are **independent** if for all x, y :

$$P(X = x, Y = y) = P(X = x)P(Y = y)$$

Setup:

- $R_X, R_Y = \{1, \dots, 6\}$
- $P(X = x) = 1/6$
- $P(Y = y) = 1/6$

Step 1: Range R_Z

- Min: $1 - 6 = -5$
- Max: $6 - 1 = 5$

$$R_Z = \{-5, \dots, 0, \dots, 5\}$$

Exercise 3: How to Calculate $P(Z = k)$

Fundamental Probability: Since X, Y are independent, each joint outcome (x, y) has:

$$P(X = x, Y = y) = \frac{1}{6} \times \frac{1}{6} = \frac{1}{36}$$

Example: $P(Z = -4)$

Find pairs (x, y) s.t. $x - y = -4$:

- Pairs: $\{(1, 5), (2, 6)\}$

$$\begin{aligned} P(Z = -4) &= \frac{1}{36} + \frac{1}{36} \\ &= \frac{2}{36} = \frac{1}{18} \end{aligned}$$

Example: $P(Z = 0)$

Find pairs (x, y) s.t. $x - y = 0$:

- Pairs: $\{(1, 1), \dots, (6, 6)\}$

$$\begin{aligned} P(Z = 0) &= 6 \times \frac{1}{36} \\ &= \frac{1}{6} \end{aligned}$$

Tip: Count the number of valid pairs and multiply by $1/36$.

Exercise 3: Final PMF

Observation: Symmetry

By symmetry, $P(Z = k) = P(Z = -k)$.

Example: $P(X - Y = 1)$ has the same number of pairs as $P(Y - X = 1)$.

- $P_Z(0) = 6/36 = 1/6$
- $P_Z(\pm 1) = 5/36$
- $P_Z(\pm 2) = 4/36 = 1/9$
- $P_Z(\pm 3) = 3/36 = 1/12$
- $P_Z(\pm 4) = 2/36 = 1/18$
- $P_Z(\pm 5) = 1/36$

Check: Does the sum equal 1?
 $(1 + 2 + 3 + 4 + 5) \times 2 + 6 = 36$. Yes!

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Exercise 4: Problem

4. Let X and Y be two independent discrete random variables with the following PMFs:

$$P_X(k) = \begin{cases} \frac{1}{4} & \text{for } k = 1 \\ \frac{1}{8} & \text{for } k = 2 \\ \frac{1}{8} & \text{for } k = 3 \\ \frac{1}{2} & \text{for } k = 4 \\ 0 & \text{otherwise} \end{cases}$$

and

$$P_Y(k) = \begin{cases} \frac{1}{6} & \text{for } k = 1 \\ \frac{1}{6} & \text{for } k = 2 \\ \frac{1}{3} & \text{for } k = 3 \\ \frac{1}{3} & \text{for } k = 4 \\ 0 & \text{otherwise} \end{cases}$$

Exercise 4 a: Joint & Conditional Probability

(a) Find $P(X \leq 2 \text{ and } Y \leq 2)$

Since X, Y are **independent**:

$$\begin{aligned}P(X \leq 2, Y \leq 2) &= P(X \leq 2) \cdot P(Y \leq 2) \\&= (P_X(1) + P_X(2)) \cdot (P_Y(1) + P_Y(2)) \\&= (1/4 + 1/8) \cdot (1/6 + 1/6) = (3/8) \cdot (1/3) = \mathbf{1/8}\end{aligned}$$

Exercise 4b: OR Logic & Independence Property

(b) Find $P(X > 2 \text{ or } Y > 2)$

Recall: Inclusion-Exclusion Principle

For any two events A and B :

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

If A and B are independent, then $P(A \cap B) = P(A)P(B)$.

Step-by-step Solution:

- Let $A = \{X > 2\}$ and $B = \{Y > 2\}$.
- **Calculate $P(A)$:** $P(X = 3) + P(X = 4) = 1/8 + 1/2 = 5/8$.
- **Calculate $P(B)$:** $P(Y = 3) + P(Y = 4) = 1/3 + 1/3 = 2/3$.

Final Calculation

$$\begin{aligned} P(X > 2 \cup Y > 2) &= P(A) + P(B) - P(A)P(B) \\ &= \frac{5}{8} + \frac{2}{3} - \left(\frac{5}{8} \cdot \frac{2}{3}\right) \end{aligned}$$

Exercise 4c

(c) Find $P(X > 2 \mid Y > 2)$

Because X and Y are **independent**, knowing something about Y does **not** change the probability of X :

$$P(X > 2 \mid Y > 2) = P(X > 2) = 1/8 + 1/2 = \mathbf{5/8}$$

Recall: Law of Total Probability (Discrete)

For any event A and a partition of the sample space $\{B_k\}$:

$$P(A) = \sum_k P(A \mid B_k)P(B_k)$$

In this problem, we partition by the values of Y .

Exercise 4d

(d) Find $P(X < Y)$ Using the values of $Y \in \{1, 2, 3, 4\}$ as the partition:

$$P(X < Y) = \sum_{k=1}^4 P(X < k) \cdot P(Y = k)$$

Calculation:

- $k = 1 : P(X < 1)P(Y = 1) = 0 \cdot (1/6) = 0$
- $k = 2 : P(X < 2)P(Y = 2) = P_X(1) \cdot (1/6) = (1/4)(1/6) = 1/24$
- $k = 3 : P(X < 3)P(Y = 3) = (1/4 + 1/8) \cdot (1/3) = (3/8)(1/3) = 1/8$
- $k = 4 : P(X < 4)P(Y = 4) = (1/4 + 1/8 + 1/8) \cdot (1/3) = (1/2)(1/3) = 1/6$

Final Sum

$$P(X < Y) = 0 + \frac{1}{24} + \frac{3}{24} + \frac{4}{24} = \frac{8}{24} = \frac{1}{3}$$

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Exercise 5: Problem

5. 50 students live in a dormitory. The parking lot has the capacity for 30 cars. If each student has a car with probability $\frac{1}{2}$ (independently from other students), what is the probability that there won't be enough parking spaces for all the cars.

Background: Binomial Distribution

Definition

A random variable X follows a **Binomial Distribution** $X \sim \text{Bin}(n, p)$ if it represents the number of successes in n independent Bernoulli trials, each with success probability p .

PMF Formula

$$P(X = k) = \binom{n}{k} p^k (1 - p)^{n-k}, \quad k = 0, 1, \dots, n$$

Application to Exercise 5:

- Each student either has a car ("Success") or doesn't ("Failure").
- Success probability $p = 1/2$.
- Total number of students $n = 50$.
- Students are **independent** of each other.

Exercise 5: Identify the Problem

1. Model the Random Variable

Let X be the number of cars owned by the 50 students. Since each student represents an independent trial with success probability $1/2$:

$$X \sim \text{Binomial}\left(50, \frac{1}{2}\right)$$

2. Identify the Event

"Not enough parking spaces" means the number of cars X exceeds the capacity (30):

$$\text{Target Probability: } P(X > 30)$$

Exercise 5: Calculation

Using the PMF of the Binomial distribution:

$$\begin{aligned}P(X > 30) &= \sum_{k=31}^{50} \binom{50}{k} \left(\frac{1}{2}\right)^k \left(\frac{1}{2}\right)^{50-k} \\&= \sum_{k=31}^{50} \binom{50}{k} \left(\frac{1}{2}\right)^{50} \quad (\text{Simplifying } p^k q^{n-k}) \\&= \left(\frac{1}{2}\right)^{50} \sum_{k=31}^{50} \binom{50}{k}\end{aligned}$$